F1 Overtake Window Predictor

DEEP LEARNING • MILESTONE 1 • TEAM JANUSFORGE • 5 NOVEMBER 2026

Team: Owen, Amine, Joshua, Ayobami. **Research question.** Can a feedforward neural network predict whether a Formula 1 driver overtakes the car directly ahead in the **next 60 seconds**, using telemetry and race context from the **previous 30 seconds**? We construct driver-pair periods every 10 seconds.

17,719	665	5
labeled windows	positive windows (3.75%)	2024 F1 races

1. Data and preprocessing

Source. OpenF1 race-session data (https://openf1.org/docs/) for five 2024 Grands Prix: Bahrain, Australia, Spain, Austria, and São Paulo. Raw inputs include driver positions and intervals, laps, overtake events, pit stops, race-control messages, and car telemetry. **Example observation.** Bahrain 2024 at 15:05:42 UTC: car **18** follows car **77**, occupies position **19**, is **1.766 seconds** behind, and does not overtake within the next 60 seconds (label 0). **Preprocessing pipeline.** Reconstruct positions at 10-second intervals and pair adjacent classified positions. Join recent time gaps (at most 20 seconds old); retain gaps greater than 0 and at most 3 seconds. Exclude nearby pit events (90 seconds), yellow-flag and safety-car/VSC/red-flag periods, and windows extending beyond race finish. We then check overtakes against position changes and sustained order; remove ambiguous or contradictory cases and verify both drivers have active lap records. Target labels: A window receives label 1 if the following driver overtakes the car directly ahead within the next 60 seconds, verified using OpenF1 overtake events and sustained position changes. Otherwise, it receives label 0. Ambiguous observations are excluded.

Features. Input features: We extract 16 numerical features: gap to the car ahead (1), race position (1), mean and latest speed of both cars (4), mean and latest relative speed (2), mean throttle (2), brake-use fraction (2), mean RPM (2), and DRS-active fraction (2). Telemetry features summarize the preceding 30 seconds. All inputs use historical information only, and the final dataset contains no missing values.

Race-separated dataset split

Partition	Races	Windows	Positive
Train	Bahrain, Spain, Austria	11,095	497
Validation	Australia	2,733	62
Test	São Paulo	3,891	106

All windows from any one race remain in a single partition, avoiding leakage from overlapping snapshots between train and test. Test data was not used to select parameters or decision thresholds.

2. Models and optimization

Baseline. Majority class (label 0): always predict *no overtake*; **0 trainable parameters**. Feedforward Neural Network (FNN): Our architecture is $16 \to 32 \to 16 \to 1$: 16 input features, two hidden layers with 32 and 16 neurons, ReLU activations, dropout of 0.15 after the first hidden layer, and one output logit. The model has 1,089 trainable parameters: $(16 \times 32 + 32) + (32 \times 16 + 16) + (16 + 1) = 1{,}089$. Features are standardized using training-set statistics only. Training uses "Adam", learning rate 0.001, batch size 256, class-weighted binary cross-entropy (positive weight 21.32), up to 60 epochs and early stopping with patience 10. Hyperparameters and model selection: We chose a compact 32/16-neuron architecture because our training set contains only 497 positive windows. Dropout 0.15 provides moderate regularization. Adam with learning rate 0.001 and batch size 256 provides a standard, efficient starting configuration. We used class-weighted binary cross-entropy, with positive weight 21.32 (10,598 negatives / 497 positives), to address class imbalance. Training ran for up to 60 epochs, with early stopping (patience 10) based on validation average precision. The best checkpoint occurred at epoch 13 (validation AP 0.1809). We selected the 0.8609 classification threshold using validation F1. Both selections used Australia only, leaving São Paulo unseen until testing. No extensive hyperparameter search was performed.

Training and held-out results

MODEL SELECTION ON AUSTRALIA • FINAL EVALUATION ON SÃO PAULO

3. Optimization evidence: training curves

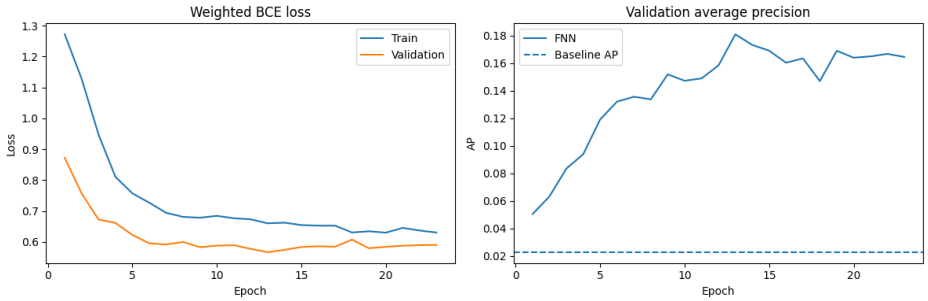


Figure 1: Training and validation weighted BCE loss and validation average precision across epochs. Validation AP peaked at 0.1809 at epoch 13. Early stopping restored the best checkpoint.

Training loss declines; validation AP reaches 0.1809 at epoch 13. Early stopping restores the best checkpoint.

4. Error metrics and final test results

Primary metric: average precision (AP, the area summary of the precision–recall curve), appropriate when only 2.72% of test windows are positive. Precision measures correctness among predicted overtakes; recall measures detected actual positives; F1 is their harmonic mean.

Model and set	Accuracy	Precision	Recall	F1	AP (PR-AUC)
Baseline / validation	0.9773	0	0	0	0.0227
FNN / validation	0.9623	0.2644	0.3710	0.3087	0.1809
Baseline / test	0.9728	0	0	0	0.0272
FNN / test	0.9465	0.1645	0.2358	0.1938	0.1148

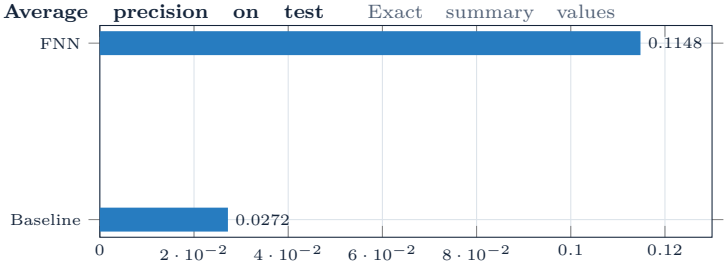
Performance:. The FNN improves test average precision from 0.0272 to 0.1148, approximately 4.22 times the majority-class baseline. Although the baseline achieves higher accuracy (97.28% versus 94.65%), it cannot detect any positive windows. The FNN correctly identifies 25 of 106 positive test windows. Validation F1 decreases from 0.3087 to 0.1938 on São Paulo, suggesting limited generalization across races. Differences in weather, race conditions, and the small validation-positive sample may contribute, although further evaluation is needed.

Test confusion matrix Rows: actual; columns: predicted

Actual	No overtake	Overtake
No overtake	**3,658 TN**	127 FP
Overtake	81 FN	**25 TP**

The model predicts 152 positive windows, of which 25 are correct, and misses 81 of the 106 actual positive windows.

Average precision on test Exact summary values



Error analysis, limitations and next steps

HELD-OUT TEST RACE • SÃO PAULO 2024

5. Error analysis

Gap to car ahead	TP	FN	FP	TN
0–0.5 s	23	13	93	120
0.5–1 s	2	44	34	1,407
1–2 s	0	19	0	1,287
2–3 s	0	5	0	844

Pattern. The model finds **23 of its 25 true positives** when the following car is within 0.5 seconds. But the same band also accounts for **93 of 127 false alarms**. Closing speed and proximity are suggestive, but not sufficient for perfect accuracy. The model misses 44 positive windows at gaps of 0.5–1 seconds and detects none above 1 second, so it is better at identifying imminent passing opportunities than predicting them from farther away. **Concrete errors.** At 16:12:07 UTC in São Paulo, car 27 followed car 14 by 0.466 s with a mean relative speed of +23.82 km/h. Its model score was 0.9976, yet the observed window was negative (false positive). At 16:04:37 UTC, car 1 followed car 30 by 0.468 s; a positive window received a score of 0.8605, just below the validation threshold 0.8609 (false negative). The examples show both false-alarm risk and threshold sensitivity. Simple comparison: To predict an overtake whenever the gap is at most 0.5 seconds achieves precision 0.145, recall 0.340, and F1 0.203 on the São Paulo test set, compared with the FNN's F1 of 0.194. This suggests proximity is a strong predictive signal and that the FNN has not yet clearly surpassed simple gap-based rules on F1. Because this comparison was examined after viewing the test results, it is not treated as a formally selected baseline.

Known limitations

- **Limited diversity:** Only five 2024 races; São Paulo involved unusual interruptions and weather.
- **Overlapping windows:** The 665 positive rows are not 665 distinct overtakes; scores are window-level, not maneuver-level.
- **Provisional labels:** Race event and position histories can be incomplete or noisy; uncertainty filters may remove genuine passes.
- **Missing context:** Tyre age/compound, weather, circuit geometry, traffic density, and gap trends are not included in the current features. Some of these variables are already available in the raw data and can be incorporated in future models.
- **Uncalibrated scores:** Class weighting helps training, but sigmoid outputs should not be interpreted as verified probabilities.

6. Work planned over the next three weeks

Week 1 – label quality and data: Manually audit recorded passes, add more races, review neutralized intervals, and report event-level metrics.
Week 2 – features and architectures: Add tyre age/compound, weather, and gap-rate features. Compare the FNN against a GRU trained on 30-second telemetry sequences, using the same race-separated validation protocol.
Week 3 – robustness and demonstration: Evaluate across multiple held-out races, compare against validation-selected gap-based baselines, investigate threshold trade-offs, calibrate predictions, and create a benchmark dashboard for faster experimentation.

Reproducibility

Source repository: github.com/amineelmess/f1-overtake-window-predictor-JanusForge . The complete experiment is available on the link above. The notebook installs its dependencies, prepares the OpenF1 dataset, trains the majority-class baseline and FNN, and evaluates both models on the same held-out race. The repository includes preprocess_all_races.py and finalize_dataset.py for reproducible preprocessing.